Dynamic Cloud-Edge-End Collaboration Reconfiguration for Federated Learning under Resource and Privacy Constraints

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Abstract

Hierarchical federated learning enables cloud-edge-end collaborative training, but fixed execution and aggregation schemes are insufficient to adapt to heterogeneous and dynamic conditions. To address this challenge, a dynamic collaboration reconfiguration framework is proposed. The framework selects collaboration modes according to the current system state and considers execution paths and aggregation strategies. Moreover, a joint learning error model is developed to characterize the effects of incomplete global fusion and update perturbation. The estimated learning error and system latency are jointly considered as optimization objectives. Feasible collaboration profiles are filtered by Pareto dominance and selected using the Chebyshev ideal point criterion. Experiments on CIFAR-10 with ResNet-18 and Fashion-MNIST with LeNet-5 evaluate dynamic collaboration under heterogeneous resource and data conditions.

Keywords: Internet of Things, hierarchical federated learning, cloud-edge-end collaboration, dynamic collaboration reconfiguration, differential privacy, device heterogeneity, dynamic privacy calibration

1. Introduction

Federated learning (FL) enables multiple end devices to collaboratively train a shared model without transferring raw data to a central server [**? ?**]. In AIoT systems, however, participating devices differ substantially in computational capability, memory capacity, communication quality, and local data distribution. Cloud-edge-end hierarchical federated learning (HFL) introduces edge servers between end devices and the cloud, providing intermediate aggregation and computation capabilities while reducing direct communication with the cloud [**? ? ?**].

Data and system heterogeneity can limit the effectiveness of fixed cloud-edge-end collaboration schemes across clients and training rounds. First, non-IID data can cause local updates to drift in different directions and can degrade simple weighted aggregation [**? ? ? ? ?**]. Second, system capabilities are heterogeneous. A weak end device may be unable to handle the same workload or meet the same deadline as a stronger device, whereas an edge or cloud server can undertake more computation [**? ?**]. A mode that suits one client in one round may be inefficient or infeasible for the same client in another round, or for another client.

Existing work on HFL and cloud-edge-end collaboration has covered personalization, resource allocation, wireless scheduling, heterogeneous models, and split training [**? ? ? ? ? ? ? ?**]. These studies address different aspects of collaborative training. However, dynamically selecting execution and aggregation configurations according to individual client capabilities and changing system conditions remains an important problem.

Different collaboration modes may expose different intermediate information, which affects the choice of protection mechanisms. In this work, differential privacy (DP) [**? ? ?**] is considered for visible updates, while homomorphic encryption (HE) [**? ?**] is considered for encrypted aggregation.

Changing a collaboration mode affects computation placement, communication paths, aggregation, and the resulting global update. These changes influence system latency and learning behavior. To evaluate their effects, system latency and a joint learning error are considered. The joint learning error accounts for algorithmic deviation, aggregation deviation and update perturbation. Resource and protection requirements are treated as feasibility constraints.

This paper studies reconfiguration of cloud-edge-end federated learning at each selection round under heterogeneous resources and changing system conditions. For each client, the system evaluates feasible collaboration and protection profiles under current conditions rather than using one fixed workflow for all clients. The main contributions are as follows.

- A dynamic collaboration configuration space for cloud-edge-end federated learning is formulated, which enables execution and aggregation configurations to be selected according to current system conditions.
- A joint learning error model is constructed, which accounts for algorithmic deviation, aggregation deviation, and update perturbation. Together with system latency, the estimated learning error forms an optimization problem with two objectives under resource and protection constraints. Feasible configurations are compared using Pareto dominance, and the final configuration is selected

using the Chebyshev ideal point criterion.

The remainder of this paper is organized as follows. Section **??** reviews related studies on hierarchical federated learning and cloud-edge-end collaboration. Section **??** introduces the system model and formulates the latency and learning error objectives under resource and privacy constraints. Section **??** presents the proposed dynamic collaboration selection algorithm. Section **??** evaluates its performance through experiments. Finally, Section **??** concludes the paper.

2. Related Work

2.1. HFL Under Heterogeneous Environments

Existing HFL studies use edge servers for intermediate aggregation between end devices and the cloud [**?**]. Research on resource management in FL has investigated model pruning under bandwidth and device constraints [**?**], adaptive local update frequency and model compression [**?** **?**], and wireless resource allocation [**?**]. Personalized FL and HFL studies have addressed data heterogeneity through client clustering, shared representations, and personalized model components [**?** **?** **?**]. These studies address different aspects of heterogeneous federated training, including resource management, communication efficiency, and model personalization.

Beyond these aspects, heterogeneous device capabilities and changing system conditions can also affect how computation and aggregation are organized across the cloud, edge, and end layers. In this work, a collaboration configuration specifies where model computation is performed, how updates are transmitted, and where aggregation takes place. Different clients may use different configurations, and their configurations may be reconsidered as system conditions change. Since frequent reevaluation incurs computational overhead, we use a configurable selection period to determine when configurations are reevaluated. Related studies have also considered the cost of frequent changes in FL scheduling decisions and reduced such overhead by maintaining selected client sets over multiple rounds [**?** **?**].

2.2. Cloud-Edge-End Collaborative Federated Learning

Cloud-edge-end collaborative FL distributes computation, communication, and aggregation tasks across end devices, edge servers, and the cloud. Different studies have explored how these tasks can be organized. FedAgg [**?**] employs bridge samples and online distillation to support collaboration among models of different sizes across the three layers. SFETEC [**?**] investigates split federated learning in Thing Edge Cloud environments, allowing model computation to be distributed across layers. Related task offloading research has also considered federated learning, privacy protection, and task efficiency in cloud-edge-end systems [**?**]. While these studies investigate different collaborative mechanisms, this work focuses on selecting feasible execution and aggregation configurations for individual clients under changing system conditions.

Task migration in mobile edge computing provides a related perspective on adapting execution placement to changing system conditions. Zhang and Zheng [**?**] investigate task migration decisions under user mobility, while Zhu et al. [**?**] study load aware task migration for balancing workloads across edge nodes. Although task migration differs from federated learning collaboration reconfiguration, these studies illustrate the importance of adapting execution placement to system conditions.

2.3. Privacy Preserving HFL

Model parameters, gradients, and intermediate representations in federated training may leak information about private data [**?**]. Differential privacy (DP) and homomorphic encryption (HE) are widely used privacy protection techniques in FL [**?** **?**]. For HFL, group local DP provides personalized privacy through sampling, randomized response, and shuffling [**?**]. Adaptive DP FL studies have explored dynamic privacy budget allocation, adaptive clipping, and client selection for adjusting privacy protection during training [**?**]. Other HFL studies combine DP based protection and CKKS based HE for privacy preserved data sharing scenarios [**?**]. Related fog FL studies combine hierarchical aggregation, personalization, and DP for IoT intrusion detection [**?**]. These studies mainly focus on the security guarantee, budget allocation, or accuracy degradation of a particular privacy mechanism. These studies mainly focus on the design, analysis, or adaptation of individual privacy mechanisms. In this work, privacy requirements are considered jointly with execution and aggregation configuration selection in cloud-edge-end federated learning.

This paper jointly considers execution, aggregation, and protection configuration in federated learning. Existing categories optimize selected resources, training steps, or privacy parameters within a prescribed collaboration structure. The proposed collaboration space enables different execution locations, aggregation paths, transmitted objects, and privacy mechanisms to be compared within one feasible decision model.

3. Model Construction

3.1. Collaborative Mode and Transmission Modeling

We consider a three-tier federated learning system consisting of N clients, a set of edge servers, and a central cloud server. Each client i is associated with an edge server $e(i)$. At round t, the system assigns a collaboration mode $h_i^t \in \mathcal{H}$ to each participating client, allowing training and aggregation to be organized across different tiers.

Collaboration interfaces. As shown in Fig. **??**, based on the information of where communication occurs across tiers in training process, three collaboration interfaces are defined. Interface I supports split training, where the Local tier exchanges intermediate information with the Edge or Cloud tier during training. Interface II submits a model update to the Edge or Cloud for aggregation after local training. Interface III submits an aggregated Edge update to the Cloud after Edge aggregation. Let L, E and C denote the Local, Edge and Cloud, respectively.

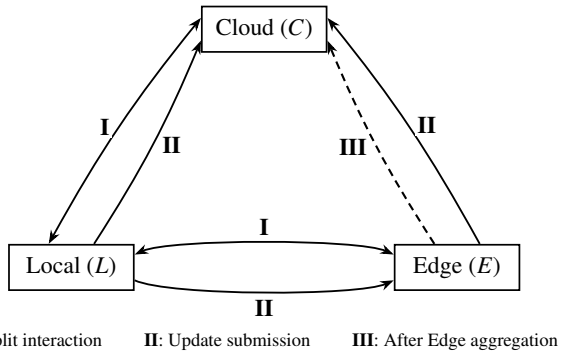


Figure 1: Collaboration interfaces and their physical positions in the cloud-edge-end architecture.

Let I, II and III indicate three interfaces. Seven client-level collaboration models are considered in this study and they are collected in the set $\mathcal{H}$:

$$\mathcal{H} = \{\text{LIE}, \text{LIIE}, \text{LIC}, \text{LIIC}, \text{LIEIIC}, \text{LIEIIIC}, \text{LIIEIIIC}\}. \quad (1)$$

For instance, LIE (L-I-E) enables split training between the Local and Edge tiers, whereas LIEIIC (L-I-E-II-C) additionally forwards the resulting update from the Edge to the Cloud through Interface II. In LIEIIIC (L-I-E-III-C), split training and Edge aggregation are repeated before the aggregated Edge update is transmitted to the Cloud through Interface III.

Under split training modes, the model is partitioned across the participating tiers, with $M_L^{h_i^t}$, $M_E^{h_i^t}$, and $M_C^{h_i^t}$ denoting the sets of model parameters assigned to the Local, Edge, and Cloud tiers, respectively. Here, $|M_k^{h_i^t}|$ represents the number of model parameters deployed at tier $k \in \{L, E, C\}$. The relative parameter counts characterize the model placement distribution, whose variation across consecutive rounds is used to approximate the redistribution overhead caused by collaboration mode switching.

In contrast, local training modes perform model training on the client before transmitting the resulting update for aggregation. Split execution is admitted only when the participating remote training tier is trusted and satisfies the required resource constraints. Under this trust assumption, intermediate activations and backward gradients exchanged with that tier do not require additional DP perturbation. In all seven modes, raw training data remain on the client.

3.1.1. Collaborative Link Blocks and Mode Flows

We characterize each collaboration mode using two basic training blocks. Let L_B be the number of training steps in each block. The split training block $S_{Lr}(L_B)$, where $r \in \{E, C\}$, represents L_B training steps performed jointly by the Local tier and the Edge or Cloud tier through split learning. The local training block $T_L(L_B)$ represents L_B training steps performed entirely at the Local tier, followed by the generation of a model update for aggregation.

The collaboration flow of each mode specifies the sequence of training, update transmission, and aggregation. Let A_E and A_C denote Edge and Cloud aggregation, respectively. For hierarchical modes, E_B denotes the number of Edge-level training and aggregation stages performed before the aggregated update is forwarded to the Cloud.

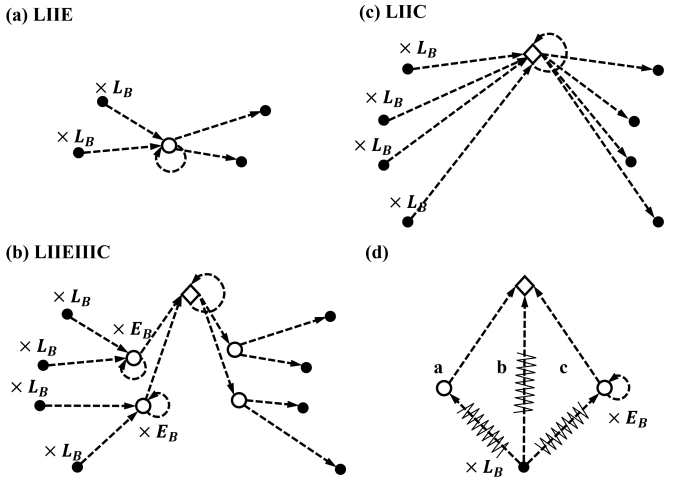


Figure 2: Execution structures of the seven collaboration modes in the cloud-edge-end architecture.

With these definitions, seven mode flows can be represented by the following procedures, where "→" is used to denote direction of data transmition, "upd" means the data updating process is involved in data transmition:

$$\begin{aligned}
\text{LIE}: &\ S_{LE}(L_B) \to A_E, \\
\text{LIIE}: &\ T_L(L_B) \xrightarrow{\text{upd}} A_E, \\
\text{LIC}: &\ S_{LC}(L_B) \to A_C, \\
\text{LIIC}: &\ T_L(L_B) \xrightarrow{\text{upd}} A_C, \\
\text{LIEIIC}: &\ S_{LE}(L_B) \xrightarrow{\text{upd}} A_C, \\
\text{LIEIIIC}: &\ [S_{LE}(L_B) \to A_E]^{E_B} \xrightarrow{\text{upd}} A_C, \\
\text{LIIEIIIC}: &\ [T_L(L_B) \xrightarrow{\text{upd}} A_E]^{E_B} \xrightarrow{\text{upd}} A_C.
\end{aligned} \quad (2)$$

Figure **??** illustrates the training and aggregation processes of the seven collaboration modes. Filled circles, hollow circles, and hollow diamonds represent the Local, Edge, and Cloud tiers, respectively. Solid arrows and hatched regions indicate training interactions under split training modes, while dashed arrows represent the transmission of model updates and the return of model parameters. Circular arrows represent aggregation operations. The notation L_B denotes the number of training steps in each training block, while E_B denotes the number of repeated training and aggregation stages at the Edge in LIEIIIC and LIIEIIIC. These processes are used to define the transmission set $\mathcal{X}_i^t(h)$ introduced below.

3.2. Lifetime Privacy Accounting

Sample-level differential privacy [**?**] is enforced through Gaussian mechanisms [**?**] during private training and on protected cross-tier releases, as required by the selected collaboration mode. For each client, Rényi differential privacy (RDP) [**?**] is used to accumulate privacy loss across training rounds. Let

$\mathcal{A}$ denote the set of Rényi orders [**?**]. For a Gaussian mechanism with noise multiplier σ, the RDP cost at order $\alpha \in \mathcal{A}$ is $\rho(\alpha, \sigma) = \alpha/(2\sigma^2)$.

Let m_i^t denote the number of Gaussian mechanisms applied to client i in round t, with corresponding noise multipliers $\sigma_{i,k}^t$. The accumulated RDP cost can be calculated as

$$\Phi_i^{t+1}(\alpha) = \Phi_i^t(\alpha) + \sum_{k=1}^{m_i^t} \rho(\alpha, \sigma_{i,k}^t). \tag{3}$$

For a given privacy parameter δ, the corresponding privacy loss is

$$\epsilon_i^t = \min_{\alpha \in \mathcal{A}} \left[\Phi_i^t(\alpha) + \frac{\log(1/\delta)}{\alpha - 1} \right]. \tag{4}$$

Let $\bar{\epsilon}$ denote the lifetime privacy budget of each client. The accumulated privacy loss must satisfy $\epsilon_i^t \le \bar{\epsilon}$ throughout training process.

For client i and candidate mode h, let $N_i^t(h)$ denote the approximate number of Gaussian mechanisms over the remaining training rounds, based on the selected mode flow and the planned Gaussian operations in its minibatch training schedule. Let R denote the total number of training rounds. For example, in LIIEIIIC mode, with one DP-SGD mechanism per training step.

For $N_i^t(h) > 0$, the noise multiplier $\sigma_i^t(h)$ is calibrated to satisfy

$$\min_{\alpha \in \mathcal{A}} \left[\Phi_i^t(\alpha) + \frac{N_i^t(h)\alpha}{2[\sigma_i^t(h)]^2} + \frac{\log(1/\delta)}{\alpha - 1} \right] \le \bar{\epsilon}. \tag{5}$$

The smallest feasible multiplier is selected. If calibration is infeasible, the candidate mode is excluded. No calibration is required when $N_i^t(h) = 0$.

For Edge group g, the noise multiplier is defined as

$$\sigma_g^t = \max_{i \in \mathcal{G}_g} \sigma_i^t(h_i^t) \tag{6}$$

Before execution, the planned privacy cost is checked against the remaining budget. The privacy ledger is updated using the executed Gaussian mechanisms, and the resulting RDP state is used for mode selection in the next round.

3.3. Joint Cost and Utility Modeling

Each collaboration mode is evaluated in terms of resource feasibility, round latency, and learning error, subject to the lifetime privacy budget in Sec. **??**.

3.3.1. End Side Resource Feasibility

Let $W_{i,L}^{\text{comp}}(h_i)$ and $M_{i,L}^{\text{mem}}(h_i)$ denote the computation and memory requirements of device i under mode h_i, and let $Q_{\text{comp},i}$ and $Q_{\text{mem},i}$ denote its available capacities. The resource feasible mode set is

$$\mathcal{H}_i^{\text{res}} = \left\{ h_i \in \mathcal{H} \mid W_{i,L}^{\text{comp}}(h_i) \le Q_{\text{comp},i},\ M_{i,L}^{\text{mem}}(h_i) \le Q_{\text{mem},i} \right\}. \tag{7}$$

If any full local update mode is feasible under the resource constraints, the candidate set is restricted to such modes. Otherwise, split modes are considered only when the participating remote training tier is trusted and has sufficient resources. All candidate modes must also satisfy the lifetime privacy constraint.

3.3.2. Path and Round Latency on a Common Timeline

As in Fig **??**, the execution, transmission, and aggregation events of the selected collaboration modes on a common timeline are used to determine their arrival times and the resulting round latency.

Transmission latency. For a payload b transmitted from tier u to tier v, let $S_{u,v,b}^t(h_i)$ denote the original payload size under collaboration mode h_i. Let $B_{i,u,v}^t$ denote the transmission rate of the link between tiers u and v for client i in round t, and let $D_{u,v}^0$ denote the fixed transmission delay of this link.

$$T_{u,v,b}^{\text{link},t}(h_i) = \frac{\kappa\left(\pi_i^{t,h_i}(u,v,b)\right) S_{u,v,b}^t(h_i)}{B_{i,u,v}^t} + D_{u,v}^0 + T_{u,v,b}^{\text{priv},t}\left(\pi_i^{t,h_i}(u,v,b)\right). \tag{8}$$

Here, $\kappa(\cdot)$ denotes the payload expansion factor, and $T_{u,v,b}^{\text{priv},t}(\cdot)$ denotes the protection processing time. For unprotected transmissions, $\kappa(\text{none}) = 1$ and $T_{u,v,b}^{\text{priv},t}(\text{none}) = 0$.

Execution latency. Let $\tau_{i,L}^t(h_i)$ and $\tau_{i,E}^t(h_i)$ denote the estimated execution times per training step at the Local and Edge tiers, respectively. For Local–Edge split modes, the latency model assumes pipelined execution across L_B training steps, giving

$$T_i^{\text{stage},t}(h_i) = L_B \max\left\{\tau_{i,L}^t(h_i), \tau_{i,E}^t(h_i)\right\} + \min\left\{\tau_{i,L}^t(h_i), \tau_{i,E}^t(h_i)\right\}. \tag{9}$$

For full local training, $T_i^{\text{stage},t}(h_i) = L_B \tau_{i,L}^t(h_i)$. For modes with repeated training stages, the corresponding stage duration is multiplied by E_B.

Reconfiguration latency. Changing the collaboration mode may require redistributing model components across the Local, Edge, and Cloud tiers. We model the resulting reconfiguration delay as

$$T_i^{\text{sw},t} = \eta_h \mathbb{I}(h_i^t \ne h_i^{t-1}) + \eta_p \frac{\|\mathbf{p}(h_i^t) - \mathbf{p}(h_i^{t-1})\|_1}{2}, \tag{10}$$

where η_h and η_p are the coefficients for the mode change cost and the placement change cost, respectively [**? ?**], and $\mathbb{I}(\cdot)$ denotes the indicator function. The normalized model placement is

$$\mathbf{p}(h) = \frac{\left[|M_L^h|, |M_E^h|, |M_C^h|\right]}{|M_L^h| + |M_E^h| + |M_C^h|}. \tag{11}$$

Path arrival and aggregation. For each candidate mode h_i, the mode flow determines the sequence of execution stages and payload transmissions leading to the relevant aggregation point. We calculate update arrival times by following this sequence and accounting for execution, transmission, and reconfiguration delays in their order of occurrence. Let $a_{i,E}^t$ and $a_{i,C}^t$ denote the

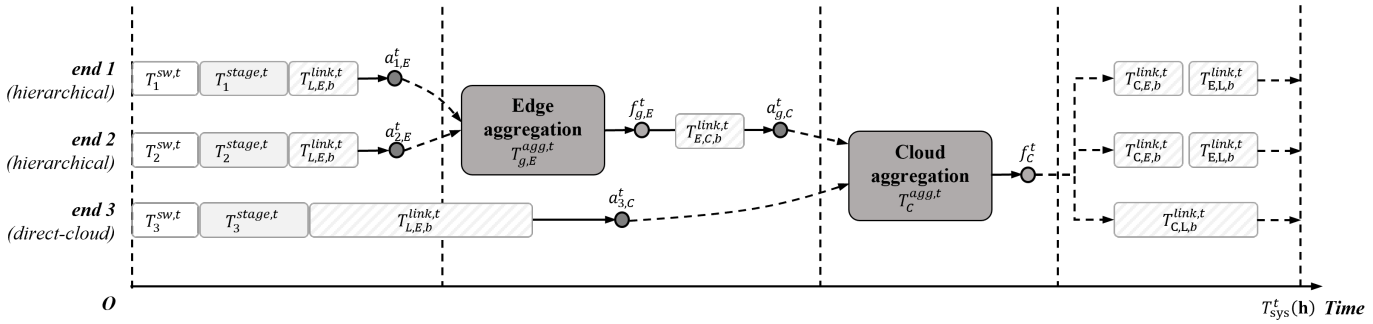


Figure 3: Illustrative common timeline for hierarchical and direct-cloud collaboration paths. The Cloud round latency includes Cloud aggregation and its required return transmissions.

arrival times of client i's update at the Edge and Cloud, respectively, when the corresponding aggregation point is involved in the selected mode.

Given a selected mode profile $\mathbf{h}^t = (h_1^t, \ldots, h_N^t)$, for an Edge aggregation group g, let $\mathcal{G}_g$ denote its participating clients. Edge aggregation begins after all required client updates have arrived. Its completion time is

$$f_{g,E}^t = \max_{i \in \mathcal{G}_g} a_{i,E}^t + T_{g,E}^{\mathrm{agg},t}, \tag{12}$$

where $T_{g,E}^{\mathrm{agg},t}$ is the Edge aggregation duration. Updates arriving earlier remain buffered until aggregation starts.

For modes that forward an Edge aggregation result to the Cloud, transmission begins after the Edge aggregation completes at time $f_{g,E}^t$. Let $a_{g,C}^t$ denote the arrival time of this result at the Cloud, including the Edge to Cloud transmission latency. Modes that transmit updates directly to the Cloud instead produce client arrival times $a_{i,C}^t$. Cloud aggregation waits for all required updates from the applicable paths. Its completion time is

$$f_C^t = \max\left\{\max_i a_{i,C}^t,\ \max_g a_{g,C}^t\right\} + T_C^{\mathrm{agg},t}, \tag{13}$$

where the two maxima are taken over the direct Cloud clients and the Edge groups participating in the current Cloud aggregation, respectively.

As illustrated in Fig. **??**, the system round latency is measured with respect to the Cloud aggregation round. Edge-only paths do not introduce Cloud synchronization dependencies. When required, return transmissions begin after Cloud aggregation completes. The system round latency is therefore

$$\begin{aligned} T_{\mathrm{sys}}(\mathbf{h}^t) = f_C^t + \max\Bigg\{ &\max_{i \in \mathcal{H}^t}\left(T_{C,E,b_i}^{\mathrm{link},t}(h_i^t) + T_{E,L,b_i}^{\mathrm{link},t}(h_i^t)\right), \\ &\max_{i \in \mathcal{D}^t} T_{C,L,b_i}^{\mathrm{link},t}(h_i^t)\Bigg\}. \end{aligned} \tag{14}$$

Here, $\mathcal{H}^t$ and $\mathcal{D}^t$ denote the hierarchical and direct-cloud client sets participating in Cloud aggregation, respectively. Return delays are evaluated using (**??**), with absent return paths excluded from the maximum.

3.3.3. Joint Learning Error Modeling

The learning objective measures the deviation of the actual Cloud update from an ideal gradient step update, accounting for errors arising from training, aggregation, and privacy mechanisms.

The mean and covariance of training perturbation. For client i under mode h_i^t, let $u_{i,0}^t$ and $u_{i,s}^t$ denote the updates obtained without and with privacy protection, respectively, under the same training configuration. The difference between these updates is

$$D_i^t = u_{i,s}^t - u_{i,0}^t. \tag{15}$$

By using D_i, the perturbation mean and covariance are calculated by

$$b_i^t = \mathbb{E}_t[D_i^t] \tag{16}$$

and

$$V_{ij}^t = \mathrm{Cov}_t(D_i^t, D_j^t). \tag{17}$$

The mean and covariance of aggregation perturbation. Let n_i denote the number of training samples at client i. The reference aggregation weight is defined by

$$p_i = \frac{n_i}{\sum_{j=1}^N n_j}. \tag{18}$$

For a mode profile $\mathbf{h}^t = \left(h_1^t, \ldots, h_N^t\right)$, let C^t denote the clients contributing to the Cloud update. In case that C^t is nonempty, their effective aggregation weights can be defined by

$$a_i^t = \begin{cases} \dfrac{n_i}{\sum_{j \in C^t} n_j}, & i \in C^t, \\ 0, & \text{otherwise.} \end{cases}$$

In case that C^t is empty, $a_i^t = 0$.

Let $\Delta_{F,0}^t = \sum_i p_i u_{i,0}^t$ and $\Delta_{P,0}^t = \sum_i a_i^t u_{i,0}^t$ respectively be full-participation reference and profile-specific update without privacy protection. The aggregation error can be defined by

$$e_{\mathrm{agg}}^t = \sum_i (a_i^t - p_i) u_{i,0}^t. \tag{19}$$

Let $\Delta_*^t = -\eta_t \nabla F(w^t)$ be the ideal gradient step. The clean algorithmic error is $e_{\mathrm{alg}}^t = \Delta_{F,0}^t - \Delta_*^t$. The actual Cloud update satisfies

$$\Delta_s^t = \sum_i a_i^t u_{i,s}^t = \Delta_*^t + e_{\mathrm{alg}}^t + e_{\mathrm{agg}}^t + \sum_i a_i^t D_i^t. \tag{20}$$

The aggregated perturbation mean and covariance are

$$b_s^t = \sum_i a_i^t b_i^t \tag{21}$$

and

$$V_s^t = \sum_{i,j} a_i^t a_j^t V_{ij}^t. \tag{22}$$

Therefore, the total update error as $e_{\text{upd}}^t = \Delta_s^t - \Delta_*^t$.

The mean and covariance of total update.

The conditional expected squared error is

$$J_{\text{learn}}^t(\mathbf{h}^t) = \mathbb{E}_t\|e_{\text{upd}}^t\|_2^2 = \left\|e_{\text{alg}}^t + e_{\text{agg}}^t + b_s^t\right\|_2^2 + \text{Tr}(V_s^t). \tag{23}$$

For mode selection, the learning objective in (**??**) is estimated using a calibration table constructed from paired clean and private training runs. For each client and mode, the table stores the estimated clean update $\widehat{u}_{i,0}^t$, perturbation mean $\widehat{b}_i^t$, and covariance trace $\widehat{v}_i^t$. Assuming conditionally independent client perturbations, the aggregate estimates are $\widehat{b}_s^t = \sum_i a_i^t \widehat{b}_i^t$ and $\widehat{v}_s^t = \sum_i (a_i^t)^2 \widehat{v}_i^t$. The ideal gradient step $\widehat{\Delta}_*^t$ is estimated from calibration data. The resulting learning objective $\widehat{J}_{\text{learn}}^t(\mathbf{h}^t)$ is evaluated during mode selection without rerunning training.

3.4. Global Multiobjective Collaborative Optimization

For each client, the decision variable is the collaboration mode $h_i^t \in \mathcal{F}_i^t$, where $\mathcal{F}_i^t \subseteq \mathcal{H}_i^{\text{res}} \subseteq \mathcal{H}$ contains the modes satisfying the resource constraints in (**??**), the activation rule in Sec. **??**, the exposure protection requirements, and the lifetime privacy constraint. The protection mechanisms and DP noise multipliers are determined by the candidate mode and current privacy ledger.

A complete mode profile is $\mathbf{h}^t = (h_1^t, \ldots, h_N^t)$. The shared Edge and Cloud aggregation and admission rules define the system feasible set $\mathcal{F}_{\text{sys}}^t \subseteq \prod_{i=1}^N \mathcal{F}_i^t$. An execution that is not admitted is reported as SKIP.

For each feasible profile, the flow model evaluates the round duration $T_{\text{sys}}(\mathbf{h}^t)$, while the calibration model estimates the learning error $\widehat{J}_{\text{learn}}^t(\mathbf{h}^t)$. The multiobjective optimization problem is

$$\underset{\mathbf{h}^t \in \mathcal{F}_{\text{sys}}^t}{\text{minimize}} \quad \begin{bmatrix} T_{\text{sys}}(\mathbf{h}^t) \\ \widehat{J}_{\text{learn}}^t(\mathbf{h}^t) \end{bmatrix}. \tag{24}$$

A bounded Pareto search maintains a nondominated archive $\widehat{\mathcal{H}}_{\text{P}}^t$. Let $T_{\min}^t, T_{\max}^t$ and $J_{\min}^t, J_{\max}^t$ denote the minimum and maximum objective values within this archive. The normalized objectives are

$$\bar{T}(\mathbf{h}^t) = \frac{T_{\text{sys}}(\mathbf{h}^t) - T_{\min}^t}{T_{\max}^t - T_{\min}^t + \gamma_n}, \qquad \bar{J}(\mathbf{h}^t) = \frac{\widehat{J}_{\text{learn}}^t(\mathbf{h}^t) - J_{\min}^t}{J_{\max}^t - J_{\min}^t + \gamma_n}, \tag{25}$$

where $\gamma_n > 0$ is a small normalization constant.

The Tchebycheff criterion selects the profile minimizing the largest normalized deviation:

$$\begin{aligned} D_\infty(\mathbf{h}^t) &= \max\{\bar{T}(\mathbf{h}^t), \bar{J}(\mathbf{h}^t)\}, \\ \mathbf{h}^{t\star} &\in \arg\min_{\mathbf{h}^t \in \widehat{\mathcal{H}}_{\text{P}}^t} D_\infty(\mathbf{h}^t). \end{aligned} \tag{26}$$

Table 1: Notation Used in Algorithm **??**

Symbol	Description
R	Total number of training rounds.
N	Number of clients.
S_P	Strategy update period.
$\Phi_i^t(\alpha)$	Accumulated RDP cost of client i at order α.
$Q_{\text{comp},i}$	Available computing capacity of client i.
$Q_{\text{mem},i}$	Available memory capacity of client i.
$\mathcal{H}_i^{\text{res}}$	Modes satisfying the resource constraints of client i.
$\mathcal{F}_i^t$	Feasible collaboration modes of client i.
$\mathcal{F}_{\text{sys}}^t$	Feasible system profiles.
$\mathbf{h}^{t-1,\star}$	Selected profile from the previous round.
$\mathbf{h}^{t,\star}$	Selected profile for the current round.
$\widehat{\mathcal{H}}_{\text{P}}^t$	Pareto archive in round t.
D_∞	Normalized Tchebycheff score.

Equal distances are resolved using system latency, learning error, and a stable profile identity, in that order. Real test accuracy is used only for experimental evaluation.

4. Algorithm

Algorithm **??** presents the dynamic collaborative mode selection procedure using the feasibility, privacy, and multiobjective models developed in the preceding sections.

Algorithm 1 Dynamic Collaborative Mode Selection

Require: $R, N, S_P, \{Q_{\text{comp},i}, Q_{\text{mem},i}\}_{i=1}^N, \{\Phi_i^1(\alpha)\}_{i=1}^N$
Ensure: $\mathbf{h}^{t,\star}$ and admission decisions for each round

for $t = 1, \ldots, R$ **do**
  Feasible Mode Construction
  Determine $\mathcal{H}_i^{\text{res}}$ using (**??**)
  Calibrate $\sigma_i^t(h_i)$ using (**??**)
  Construct $\mathcal{F}_i^t$ using the privacy and activation constraints
  Construct $\mathcal{F}_{\text{sys}}^t$ using the system constraints
  Reconfiguration Decision
  if $\mathcal{F}_{\text{sys}}^t = \emptyset$ **then**
    Assign SKIP to all clients
  else if $t > 1$, $\text{mod}(t-1, S_P) \neq 0$, and the previous configuration remains feasible **then**
    $\mathbf{h}^{t,\star} \leftarrow \mathbf{h}^{t-1,\star}$
  else
    Objective Evaluation
    Generate initial profiles from previous choices, latency, coverage, and learning cost
    Apply required coverage repair
    Evaluate $T_{\text{sys}}(\mathbf{h}^t)$ using (**??**)
    Estimate $\widehat{J}_{\text{learn}}^t(\mathbf{h}^t)$ using calibrated statistics and (**??**)
    Multiobjective Selection
    Initialize $\widehat{\mathcal{H}}_{\text{P}}^t$ from evaluated profiles using (**??**)
    Explore new profiles by changing one client's mode and protection settings
    Evaluate new profiles using (**??**) and update $\widehat{\mathcal{H}}_{\text{P}}^t$ by Pareto dominance
    Select $\mathbf{h}^{t,\star}$ using (**??**) and (**??**)
  end if
  Training Execution
  Apply admission adjustments according to tier capacity

Reserve authorized Sample DP costs before execution
Execute admitted clients and update the global model
Update $\{\Phi_i^{t+1}(\alpha)\}_{i=1}^{N}$ using (**??**)
Refresh the system state
end for

5. Experiments

We investigate the accuracy–time trade-off of dynamic collaboration under heterogeneous computing resources, communication conditions, and privacy budgets. The evaluation covers baseline methods, mode reconfiguration, and resource-constrained training scenarios.

5.1. Experimental Setup

The CIFAR-10 experiments adopt a pretrained ResNet-18 and freeze its feature-extraction backbone; only the 5,130 parameters in the final classification head are updated. We select 24,000 training images and 2,000 test images. The edge-label-skew partition creates non-IID label distributions across edge groups, with 20% of each client's assigned data reserved for local validation. The current 100-client, 10-edge topology assigns ten clients to each edge.

Heterogeneous resources and links. Each client has an individual compute factor and memory-capacity factor, and each edge server has its own compute factor. The heterogeneity parameters in Table **??** control the dispersion of these capacities. S1 increases both client- and edge-level heterogeneity; S2 introduces more pronounced temporal variation in network connectivity. Client–edge, client–cloud, and edge–cloud links have distinct throughput and propagation delays; network jitter and a periodic variation component change their communication conditions across training rounds.

Evaluation criteria and baselines. Our primary measure is *test accuracy as a function of cumulative logical time*, reflecting the accuracy–time trade-off under resource and privacy constraints. We additionally report final accuracy, late-stage mean accuracy, total logical time, participation rate, and selection cost.

The baselines include Fixed FedAvg, Fixed SplitFed, Fixed HFL, Random, Individual Optimal, and NSGA-II. Individual Optimal selects client modes independently, while NSGA-II uses a multiobjective population search. Within each scenario, all methods use matched model initialization, data splits, budgets, and system settings. The mode ablation changes only the reconfiguration policy.

5.2. Search Space Reduction and Selection Cost

Objective. This experiment measures the reduction in candidate collaboration profiles and the computation required to select one profile under the coupled client, edge, and cloud constraints.

Procedure. We use the first three recorded rounds of the S2 experiment to compare the combinatorial profile count with the feasible-mode search space and the number of full profiles actually evaluated. With seven supported collaboration modes and 100 clients, the unconstrained mode space contains 7^{100} profiles. In this scenario, client-level filtering retains two modes for 82 clients, four modes for eight clients, and no active mode for ten clients. The resulting Cartesian product over available active clients has $2^{82}4^8$ combinations before shared edge/cloud constraints. Bounded Pareto search evaluates only 21, 22, and 21 new complete profiles in the first three rounds, respectively.

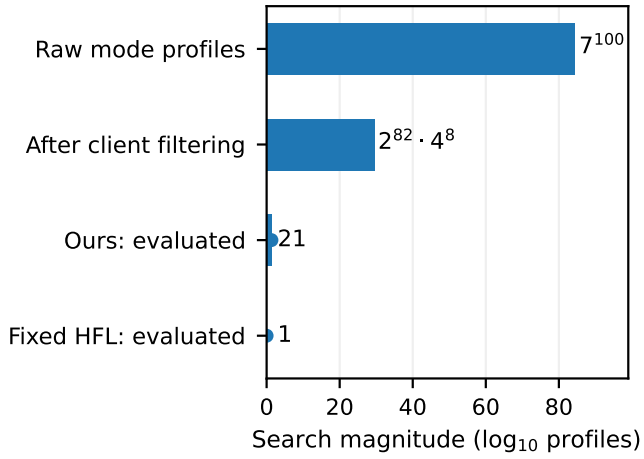


Figure 4: Search-space magnitude in the first S2 round. The first two bars represent combinatorial profile counts; the last two show actual distinct complete-profile evaluations in Ours and Fixed HFL.

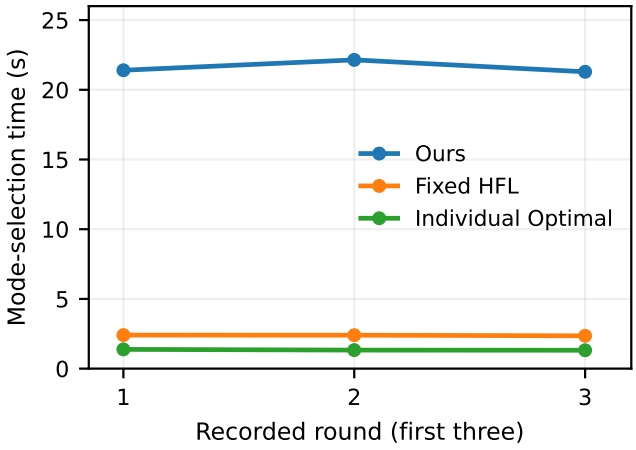


Figure 5: Measured mode-selection computation time for Ours, Fixed HFL, and Individual Optimal in the first three S2 rounds.

Results. Fig. **??** illustrates how local feasibility filtering and bounded profile exploration avoid enumerating the full combinatorial space. Over these three rounds, Ours evaluates 21.33 distinct profiles per round on average. Fig. **??** reports the corresponding selection computation time: 21.61 s for Ours, 2.39 s for Fixed HFL, and 1.34 s for Individual Optimal, averaged over the same rounds. This comparison quantifies the search-workload reduction as well as the additional computational cost of global dynamic selection.

Table 2: Experimental configurations for model training and cloud–edge–end collaboration. A dash denotes a pending Fashion-MNIST / LeNet-5 configuration.

Parameter	**CIFAR-10 / ResNet-18**		**Fashion-MNIST / LeNet-5**
	Main / S1 / S2	**Mode ablation**	**Second dataset**
Learning and privacy configuration			
Dataset / model	CIFAR-10 / pretrained ResNet-18		Fashion-MNIST / LeNet-5
Trainable parameters	5,130 (classification head only)		–
Train / test subset size	24,000 / 2,000		–
Clients / edge servers	100 / 10		–
Data partition / local validation	Edge-label skew / 20% per client		–
Local epochs / learning rate	3 / 0.01		–
Split workload batch size	128		–
Privacy budget (ϵ, δ)	$(8, 10^{-5})$, sample-level RDP		–
HE cost model / seed	CKKS-profiled / 42		–
Client, edge, and network configuration			
Clients per edge (current topology)	10		–
Client / edge heterogeneity factors	Main/S2: 2 / 1.5; S1: 4 / 3	2 / 1.5	–
Resource / memory limits	1.35 / 1.35	100 / 100	–
Edge / cloud CPU caps	Main: 100 / 200; S1/S2: 15 / 20	10,000 / 20,000	–
Link rates (C–E / C–Cloud / E–Cloud)	5 / 2.2 / 8 MB/s		–
Base link delays (same order)	15 / 40 / 10 ms		–
Jitter strength	Main: 0.25; S1: 0.50; S2: 0.45	0.25	–
Periodic network amp. / period	Main/S1: 0.20 / 3; S2: 0.55 / 12	0.20 / 3	–
Pareto archive / beam size	16 / 4	8 / 2	–
Max search iters / neighbor top-k	50 / 16	20 / 16	–
Mode reconfiguration period	$P = 1$ (Ours)	$P = 1, 5, 10$; fixed mode	–

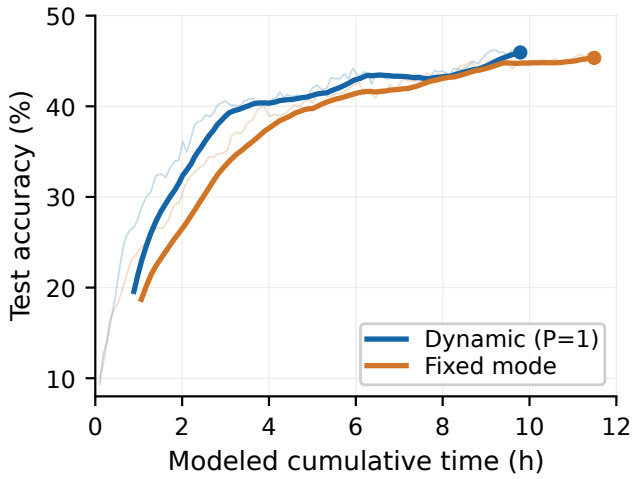


Figure 6: Mode-selection ablation: test accuracy versus cumulative logical time. The bold curves show ten-round moving averages; terminal markers indicate the final ten-round mean.

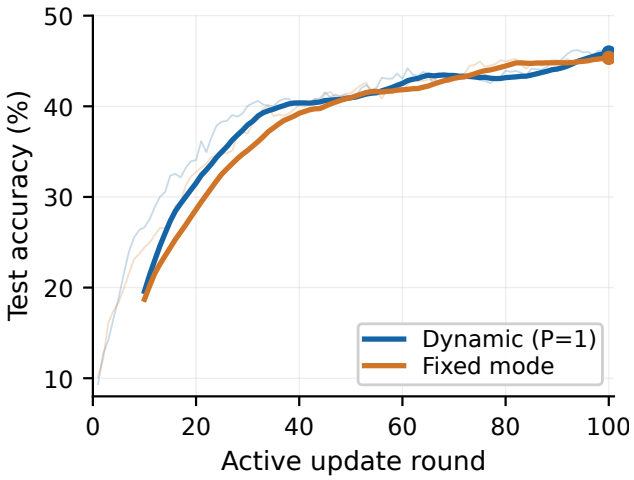


Figure 7: Accuracy against active training rounds for the same policies, with observed values and ten-round moving averages.

5.3. Ablation Study

5.3.1. Mode Selection and Reconfiguration Period

We evaluate the effect of periodically reassigning collaboration modes under the same dynamic privacy policy. Dynamic Mode updates mode assignments every P training rounds; Fixed Mode keeps its initial assignments. The paired $P = 1$ and Fixed Mode experiments use the settings in Table **??** and perform 100 active updates and 50,000 optimizer steps per policy. Periods $P = 5$ and $P = 10$ follow the same configuration.

As shown in Table **??**, $P = 1$ improves the measured final accuracy by 0.60 percentage points (1.32% relative) and reduces modeled cumulative logical time by 14.81% compared with Fixed Mode. The last-ten-active-round mean accuracy is 45.94% versus 45.34%. Figs. **??** and **??** show the accuracy–time and round-based convergence profiles. Additional reconfiguration periods will quantify the sensitivity of the accuracy–time trade-off to less frequent mode changes.

5.4. Comparison with Baselines

We compare Ours with Individual Optimal, Random, Fixed FedAvg, Fixed SplitFed, Fixed HFL, and NSGA-II under the

Table 3: Mode-selection ablation and reconfiguration periods (CIFAR-10, seed 42). Relative changes use Fixed Mode as reference; ↑ indicates higher accuracy and ↓ indicates lower modeled logical time. Last-10 is the mean over the ten final *active* rounds. Dashes indicate experiments not yet completed.

Policy	P	**Final acc. (%)**	**Δacc. (%)**	**Last-10 (%)**	**Time (s)**	**Δtime (%)**	**Active rounds**
Dynamic Mode	1	46.10	↑1.32	45.94	35,237.88	↓14.81	100
Dynamic Mode	5	–	–	–	–	–	–
Dynamic Mode	10	–	–	–	–	–	–
Fixed Mode	Frozen	45.50	Ref.	45.34	41,364.37	Ref.	100

same constrained experimental profile. The fixed-mode baselines represent the corresponding modes in our implementation rather than full external reimplementations. Table **??** reports the completed 100-round comparisons.

Among the completed baselines, Fixed HFL achieves the highest final accuracy (45.80%), followed by Individual Optimal (45.60%). Fixed SplitFed reaches 6.90% accuracy with an average of 40 participating clients under its assigned execution profile.

5.5. Discussion

The constrained baseline comparison and mode-selection ablation examine complementary aspects of system efficiency. Under the constrained main configuration, Ours achieves 45.00% test accuracy, compared with 45.80% for Fixed HFL and 45.60% for Individual Optimal. Under the matched mode-ablation configuration, dynamic reconfiguration ($P = 1$) raises test accuracy by 0.60 percentage points while reducing cumulative logical time by 14.81% relative to fixed assignment. Subsequent controlled experiments examine hardware heterogeneity, edge placement, network variability, resource budgets, dataset scaling, and switching periods.

6. Conclusion

This paper presented a dynamic cloud-edge-end collaboration reconfiguration framework for federated learning under resource and privacy constraints. The unified collaboration space represents execution placement, aggregation topology, transmitted objects, and return paths as the dynamic decision, while a mode-specific exposure contract determines where DP and HE are required. The joint model evaluates latency, communication, candidate-specific DP distortion, encrypted aggregation overhead, and estimated convergence error cost. The constrained multiobjective formulation is solved by bounded Pareto search and ideal point Tchebycheff selection.

The current framework uses profiled computation and link rates in its logical latency model, while the training process and CKKS aggregation are executed on one host. Future work will combine online profiling with physical cloud-edge-end deployment and will further reduce reconfiguration overhead through parallel profile evaluation, distributed search, and warm starts.

Appendix A. Analysis of Joint Learning Error

This appendix establishes the relationship between the joint learning error in Sec. **??** and the convergence of the global model.

Appendix A.1. Aggregation Error Bound

For a fixed mode profile, the aggregation error is given by (**??**). The clean updates and aggregation weights are treated as fixed when evaluating the conditional privacy perturbation moments.

Lemma 1. *Suppose that* $\|u_{i,0}^t\|_2 \le U_i^t$ *for every client i. Then*

$$\|e_{\text{agg}}^t\|_2 \le \sum_{i=1}^{N} |a_i^t - p_i| U_i^t. \tag{A.1}$$

If $U_i^t \le U^t$ *for all clients, then* $\|e_{\text{agg}}^t\|_2 \le U^t \|\mathbf{a}^t - \mathbf{p}\|_1$.

Proof. From (**??**) and the triangle inequality,

$$\begin{aligned} \|e_{\text{agg}}^t\|_2 &= \left\| \sum_i (a_i^t - p_i) u_{i,0}^t \right\|_2 \\ &\le \sum_i |a_i^t - p_i| \|u_{i,0}^t\|_2 \\ &\le \sum_i |a_i^t - p_i| U_i^t. \end{aligned} \tag{A.2}$$

Applying the common bound U^t gives the second result. □

Appendix A.2. Joint Update Error

Let $\mathbb{E}_t$ denote conditional expectation given the round information, selected mode profile, and common training schedule. Under this conditioning, the clean reference updates and aggregation weights are fixed.

Define the aggregated trajectory difference as $D_s^t = \sum_i a_i^t D_i^t$. Using (**??**), the total update error satisfies

$$e_{\text{upd}}^t = e_{\text{alg}}^t + e_{\text{agg}}^t + D_s^t. \tag{A.3}$$

Proposition 2. *Let* $b_s^t = \mathbb{E}_t[D_s^t]$ *and* $V_s^t = \text{Cov}_t(D_s^t)$. *Then*

$$\mathbb{E}_t \|e_{\text{upd}}^t\|_2^2 = \left\| e_{\text{alg}}^t + e_{\text{agg}}^t + b_s^t \right\|_2^2 + \text{Tr}(V_s^t). \tag{A.4}$$

Proof. Write $D_s^t = b_s^t + \zeta_s^t$, where $\mathbb{E}_t[\zeta_s^t] = 0$ and $\text{Cov}_t(\zeta_s^t) = V_s^t$. From (**??**),

$$e_{\text{upd}}^t = e_{\text{alg}}^t + e_{\text{agg}}^t + b_s^t + \zeta_s^t. \tag{A.5}$$

Expanding its squared norm and taking conditional expectation eliminates the cross term involving ζ_s^t. Furthermore, $\mathbb{E}_t \|\zeta_s^t\|_2^2 = \text{Tr}(V_s^t)$. The stated identity follows. □

The covariance satisfies $V_s^t = \sum_{i,j} a_i^t a_j^t V_{ij}^t$. When the client perturbations are conditionally independent, $V_s^t = \sum_i (a_i^t)^2 V_{ii}^t$. This simplification is used by the calibration based selection objective.

Table 4: Resource-constrained CIFAR-10 comparison (seed 42).

Method	Final Acc. (%)	Last 10 (%)	Loss	Logical time (s)	Mean clients
Ours	45.00	–	1.6644	31,056	–
Individual Optimal	45.60	45.33	1.6477	25,841	90.0
Random	41.45	41.97	1.6718	55,042	90.0
Fixed FedAvg	44.15	43.69	1.6212	69,958	82.0
Fixed SplitFed	6.90	7.85	2.4631	31,300	40.0
Fixed HFL	**45.80**	45.12	1.6327	23,769	82.0
NSGA-II	43.00	43.42	1.6505	37,650	89.1

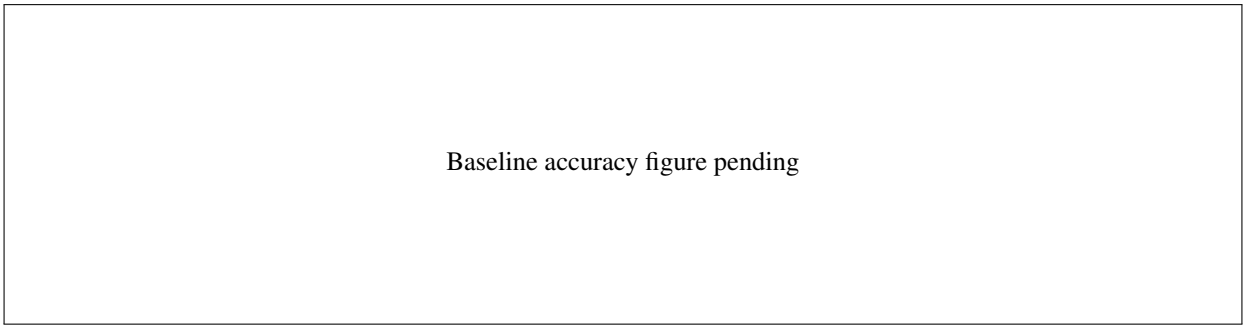


Figure 8: Test accuracy across rounds for the six plotted constrained baselines.

Appendix A.3. Expected Loss and Convergence

Assume that the global objective F is β-smooth, so that

$$F(y) \le F(x) + \langle \nabla F(x), y - x \rangle + \frac{\beta}{2}\|y - x\|_2^2. \tag{A.6}$$

Proposition 3. *Suppose that F is β-smooth and $0 < \eta_t \le 1/\beta$. For the model update $w^{t+1} = w^t + \Delta_s^t$, the conditional expected loss satisfies*

$$\mathbb{E}_t[F(w^{t+1})] \le F(w^t) - \frac{\eta_t}{2}\|\nabla F(w^t)\|_2^2 + \frac{J^t_{\text{learn}}(\mathbf{h}^t)}{2\eta_t}. \tag{A.7}$$

Proof. By the definition of the total update error,

$$w^{t+1} = w^t - \eta_t \nabla F(w^t) + e^t_{\text{upd}}. \tag{A.8}$$

Let $g^t = \nabla F(w^t)$. Applying (**??**) gives

$$\begin{aligned} F(w^{t+1}) \le F(w^t) &+ \langle g^t, -\eta_t g^t + e^t_{\text{upd}} \rangle \\ &+ \frac{\beta}{2}\| - \eta_t g^t + e^t_{\text{upd}}\|_2^2. \end{aligned} \tag{A.9}$$

Since $\beta \le 1/\eta_t$, the quadratic term is bounded by

$$\begin{aligned} \frac{\beta}{2}\| - \eta_t g^t + e^t_{\text{upd}}\|_2^2 \le \frac{\eta_t}{2}\|g^t\|_2^2 &\\ - \langle g^t, e^t_{\text{upd}} \rangle &\\ + \frac{1}{2\eta_t}\|e^t_{\text{upd}}\|_2^2. & \end{aligned} \tag{A.10}$$

Substitution cancels the inner products involving e^t_{upd}, yielding

$$F(w^{t+1}) \le F(w^t) - \frac{\eta_t}{2}\|g^t\|_2^2 + \frac{1}{2\eta_t}\|e^t_{\text{upd}}\|_2^2. \tag{A.11}$$

Taking conditional expectation and applying (**??**) completes the proof. □

The bound holds for a biased update error and does not require $\mathbb{E}_t[e^t_{\text{upd}}] = 0$.

For the convergence result, assume that F satisfies the Polyak–Lojasiewicz (PL) condition with parameter $\mu > 0$:

$$\|\nabla F(w)\|_2^2 \ge 2\mu(F(w) - F^*), \tag{A.12}$$

where $F^* = \inf_w F(w)$.

Corollary 4. *Under the assumptions of Proposition* **??** *and the PL condition,*

$$\begin{aligned} \mathbb{E}_t[F(w^{t+1}) - F^*] \le (1 - \mu\eta_t)(F(w^t) - F^*) &\\ + \frac{J^t_{\text{learn}}(\mathbf{h}^t)}{2\eta_t}. & \end{aligned} \tag{A.13}$$

Proof. Applying (**??**) to the gradient term in (**??**) gives the stated recursion. □

For a constant step size $\eta_t = \eta$ satisfying $0 < \mu\eta \le 1$, suppose that $J^t_{\text{learn}}(\mathbf{h}^t) \le \overline{J}$ for all rounds. Setting $q = 1 - \mu\eta$ and taking successive expectations yields

$$\begin{aligned} \mathbb{E}[F(w^T) - F^*] \le q^T(F(w^0) - F^*) &\\ + \frac{\overline{J}}{2\eta}\sum_{t=0}^{T-1} q^{T-1-t}. & \end{aligned} \tag{A.14}$$

Consequently,

$$\limsup_{T\to\infty} \mathbb{E}[F(w^T) - F^*] \le \frac{\overline{J}}{2\mu\eta^2}. \tag{A.15}$$

Thus, the joint learning error controls the residual term in the expected convergence bound. The selection objective $\widehat{J}^t_{\text{learn}}$ approximates this quantity using calibrated update statistics.

Pre-decision online learning-error bound.. For a candidate profile $\mathbf{h}^t$, let the client weights $a_i^t(\mathbf{h}^t)$ be determined by the predicted admission and aggregation topology. For the main derivation below, the clean reference update $u_{i,0}^t(h_i)$ is defined by running the same *clipped* training protocol with the privacy randomness removed; it is a mathematical counterfactual and is not executed when evaluating a candidate. This convention must be reconciled with the main text's phrase "without privacy protection": if the paper instead means *unclipped* SGD, the clipping-bias extension below is mandatory. The reference updates for nonadmitted clients are likewise defined counterfactually. The conditional expectation in (**??**) is taken over fresh privacy randomness, conditional on the round state, candidate profile, and common minibatch schedule.

The clean deviation from the ideal gradient step is

$$z^t(\mathbf{h}^t) = e_{\text{alg}}^t + e_{\text{agg}}^t = \sum_i a_i^t(\mathbf{h}^t) u_{i,0}^t(h_i) - \Delta_*^t. \tag{A.16}$$

Hence the exact objective is

$$J_{\text{learn}}^t(\mathbf{h}^t) = \mathbb{E}_t \left\| z^t(\mathbf{h}^t) + \sum_i a_i^t(\mathbf{h}^t) D_i^t(h_i) \right\|_2^2. \tag{A.17}$$

The following bound can be computed without paired offline clean/private training if its assumptions are enforced.

Clipped clean-update bound.. Suppose each clean-noise-free SGD step uses an average of gradients clipped at threshold C and no additional unbounded optimizer step. With $K_i^t(h)$ local steps and learning rates $\eta_{i,k}^t(h)$,

$$\left\| u_{i,0}^t(h) \right\|_2 \le U_i^t(h) := C \sum_{k=0}^{K_i^t(h)-1} \eta_{i,k}^t(h). \tag{A.18}$$

If $\|\nabla F(w^t)\|_2 \le G_t$ is a valid bound, then a profile-uniform algorithmic bound is

$$A_t := \sum_i p_i \max_{h \in \mathcal{H}_i} U_i^t(h) + \eta_t G_t \;\ge\; \|e_{\text{alg}}^t\|_2. \tag{A.19}$$

This expression is conservative. In particular, clipping private training gradients alone does not imply a bound on the unclipped $\nabla F(w^t)$ appearing in Δ_*^t. By (**??**), define

$$B_t(\mathbf{h}^t) := \sum_i \left| a_i^t(\mathbf{h}^t) - p_i \right| U_i^t(h_i). \tag{A.20}$$

Both A_t and $B_t(\mathbf{h}^t)$ are calculable before training. When all $U_i^t = U_t$ and the contributing set has total reference weight $r_t = \sum_{i \in C^t} p_i > 0$, we have the useful identity $B_t(\mathbf{h}^t) = 2U_t(1 - r_t)$. This special case is a coverage bound, not an exact characterization of aggregation-direction error.

Privacy perturbation along local SGD trajectories.. For a fixed candidate (i, h), couple a clean-noise-free trajectory $w_{i,k}^0$ and a private trajectory $w_{i,k}^s$ using the same initial point and minibatch schedule:

$$w_{i,k+1}^0 = w_{i,k}^0 - \eta_{i,k} g_{i,k}(w_{i,k}^0), \tag{A.21}$$

$$w_{i,k+1}^s = w_{i,k}^s - \eta_{i,k} [g_{i,k}(w_{i,k}^s) + \xi_{i,k}]. \tag{A.22}$$

Assume the clipped-average gradient map $g_{i,k}$ is $L_{i,k}$-Lipschitz in the trainable model parameters. Assume the fresh noise $\xi_{i,k}$ is conditionally mean-zero, independent of the preceding private iterates, and satisfies $\mathbb{E}_t\|\xi_{i,k}\|_2^2 = q_{i,k}(h)$. Let $D_{i,k} = w_{i,k}^s - w_{i,k}^0$ and set $R_{i,0}(h) = 0$. Define the deterministic recursion

$$R_{i,k+1}(h) = (1 + \eta_{i,k} L_{i,k})^2 R_{i,k}(h) + \eta_{i,k}^2 q_{i,k}(h). \tag{A.23}$$

Then $\mathbb{E}_t\|D_{i,K_i(h)}\|_2^2 \le R_{i,K_i(h)}(h)$. In particular, with Gaussian gradient noise of covariance $(\sigma_{i,k}(h)C/B_{i,k})^2 I_{d_{i,k}}$,

$$q_{i,k}(h) = d_{i,k} \left(\frac{\sigma_{i,k}(h) C}{B_{i,k}} \right)^2. \tag{A.24}$$

Here $B_{i,k}$ is the actual denominator of the clipped-average in the implemented optimizer, and the formula presumes that the multiplier and adjacency convention match the DP accountant. The dimension $d_{i,k}$ counts coordinates actually affected by that Gaussian injection. Privacy perturbations introduced on shared Edge updates or at other stages must be modeled separately; they cannot automatically be treated as independent client noises.

Proposition 5 (Online joint learning-error upper bound)**.** *Condition on the round state, profile, and common minibatch schedule. Assume the bounds in* (**??**)–(**??**) *and the second-moment recursions* (**??**) *hold for every candidate mode. Suppose, additionally, that the Cloud update has the additive client perturbation form in* (**??**)*, with no unmodeled Edge-level noise or encryption approximation error. Then*

$$\boxed{\begin{aligned} J_{\text{learn}}^t(\mathbf{h}^t) &\le \overline{J}_{\text{learn}}^t(\mathbf{h}^t) \\ &:= \left[A_t + B_t(\mathbf{h}^t) + \sum_i |a_i^t(\mathbf{h}^t)| \sqrt{R_{i,K_i(h_i)}(h_i)} \right]^2. \end{aligned}} \tag{A.25}$$

This bound requires no independence assumption among client perturbations. It bounds the original joint squared update error, including the effect of all mean and covariance cross terms, but does not claim to recover their exact signed values.

Proof. Writing $D_{i,k+1}$ from (**??**), conditional mean-zero fresh Gaussian noise eliminates the cross term between the past-adapted iterate difference and the current noise. Lipschitz continuity yields

$$\mathbb{E}_t\|D_{i,k+1}\|^2 \le (1 + \eta_{i,k} L_{i,k})^2 \mathbb{E}_t\|D_{i,k}\|^2 + \eta_{i,k}^2 q_{i,k}(h), \tag{A.26}$$

proving the recursive second-moment bound by induction. Using (**??**) and Minkowski's inequality in the L^2 space of random vectors,

$$\sqrt{J_{\text{learn}}^t(\mathbf{h}^t)} \le \|z^t(\mathbf{h}^t)\|_2 + \sum_i |a_i^t| \sqrt{\mathbb{E}_t\|D_i^t\|_2^2} \tag{A.27}$$

$$\le \|e_{\text{alg}}^t\|_2 + \|e_{\text{agg}}^t\|_2 + \sum_i |a_i^t| \sqrt{R_{i,K_i}(h_i)} \tag{A.28}$$

$$\le A_t + B_t(\mathbf{h}^t) + \sum_i |a_i^t| \sqrt{R_{i,K_i}(h_i)}. \tag{A.29}$$

Squaring the last inequality establishes the proposition. □

If the clean reference is unclipped SGD.. There are two valid conventions for the clean counterfactual, but they must not be mixed. If $u^t_{i,0}$ denotes an *unclipped* noise-free trajectory, introduce the intermediate clipped/noiseless update $u^t_{i,c}$. Then

$$D^t_i = (u^t_{i,s} - u^t_{i,c}) + (u^t_{i,c} - u^t_{i,0}). \tag{A.30}$$

The first term is controlled by $\sqrt{R_{i,K_i}}$. For the second, let $g^{\text{raw}}_{i,k}$ be the unclipped minibatch gradient and $g^c_{i,k}$ the clipped-average gradient. Suppose $\|g^c_{i,k}(x) - g^c_{i,k}(y)\| \le L_{i,k}\|x - y\|$, and the clipping discrepancy on the unclipped trajectory is $\|g^c_{i,k}(w^0_{i,k}) - g^{\text{raw}}_{i,k}(w^0_{i,k})\| \le \kappa_{i,k}$. The deterministic clipping-trajectory difference is bounded by

$$H_{i,0} = 0, \qquad H_{i,k+1} = (1 + \eta_{i,k}L_{i,k})H_{i,k} + \eta_{i,k}\kappa_{i,k}. \tag{A.31}$$

Thus replace $\sqrt{R_{i,K_i}}$ in (**??**) with $H_{i,K_i} + \sqrt{R_{i,K_i}}$. Importantly, the clean-update bound (**??**) must also be replaced by a valid bound for *unclipped* updates, for example $U_i \le \sum_k \eta_{i,k}G_{i,k}$ under the explicit assumption $\|g^{\text{raw}}_{i,k}\| \le G_{i,k}$. A bound on clipped gradient norms alone cannot certify this unclipped-clean formulation. If all per-example raw gradients have norm at most $G_{i,k}$, a conservative clipping-discrepancy bound is $\kappa_{i,k} \le (G_{i,k} - C)_+$.

Shared Edge/Cloud release noise.. The seven collaboration modes may include a Gaussian mechanism on an aggregated Edge-to-Cloud release. Such noise is shared by the clients in its group, so assigning a separate independent copy of it to every client would be incorrect. If the actual Cloud update contains an extra term $\sum_g c^t_g Z^t_g$ where each Z^t_g is an independent, zero-mean *post-training* Gaussian vector with covariance Ω^t_g, independent of client training randomness, the right-hand side of (**??**) can be tightened to

$$\overline{J}^t_{\text{learn,shared}} = \overline{J}^t_{\text{learn}} + \sum_g (c^t_g)^2 \operatorname{Tr}(\Omega^t_g). \tag{A.32}$$

If the shared releases are not independent, the universally valid Minkowski alternative is to add $\sum_g |c^t_g| \sqrt{\mathbb{E}_t\|Z^t_g\|^2}$ *inside* the square in (**??**). If an Edge-level mechanism perturbs a parameter trajectory *before* later optimization steps, its propagation must instead be included in a corresponding stability recursion. These extra mechanisms must also be counted in the appropriate privacy ledgers.

For E^t_i complete local epochs and a deterministic batch schedule, $K^t_i = E^t_i\lceil n^{\text{train}}_i / B\rceil$, where $B = 256$ is the nominal batch size in the intended BatchAccounted DP-SGD protocol. Each actual Gaussian event is included in both the noise recursion and the RDP ledger; the mechanisms must use the same event count and multiplier convention. For later epochs, n^{train}_i excludes the local validation split. Additional release mechanisms, if present, are not counted as extra SGD steps and require their own privacy and error terms.

A tighter direction-aware alternative may replace the $A_t + B_t$ term with $\|\sum_i a^t_i \widetilde{u}^t_i(h_i) - \widetilde{\Delta}^t_*\|_2 + \sum_i |a^t_i| r^t_i(h_i) + r^t_*$, provided that valid bounds $\|u^t_{i,0}(h_i) - \widetilde{u}^t_i(h_i)\| \le r^t_i(h_i)$ and $\|\Delta^t_* - \widetilde{\Delta}^t_*\| \le r^t_*$ are available. Historical DP releases may inform these estimates, but without justified error radii this is an empirical online surrogate, not a certified upper bound. DP postprocessing of already released updates incurs no new privacy event, whereas obtaining new private statistics generally does.

Connection with the descent guarantee..